Volume 03 - Book 03.11 Provider Ecosystem

Version v13 draft

README.md

Book 03.11 - Provider Ecosystem

Version: v13 draft

Status: Draft

Document ID: MAL-V03-B03-11-README

Purpose

Define the Provider Ecosystem blueprint package and its subsystem decomposition as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for the Provider Ecosystem blueprint package and its subsystem decomposition. It does not implement product code or bypass the documentation-first governance model.

Responsibilities

- Establish a canonical blueprint boundary for the Provider Ecosystem blueprint package and its subsystem decomposition.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Provider Ecosystem.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

Outputs

- Blueprint contract and subsystem boundaries for the Provider Ecosystem blueprint package and its subsystem decomposition.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.

- Acceptance evidence that downstream implementation can use before writing system code.

Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.
- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

Events

- ProviderEcosystemDefined
- ProviderEcosystemRegistered
- ProviderEcosystemStateChanged
- ProviderEcosystemReviewRequested
- ProviderEcosystemGovernanceBlocked

Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
- Retention metadata that separates draft, active, archived, deprecated, and superseded records.

Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
- Classify inputs, outputs, events, logs, metrics, memory, and exported artifacts before sharing or persistence.
- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

Metrics

- Adoption rate, successful completion rate, rejection rate, rework count, and user trust indicators.
- Latency, cost, throughput, availability, error rate, fallback rate, and recovery time.
- Quality score, benchmark pass rate, validation pass rate, review backlog, and stale-record count.
- Security denial count, governance exception count, policy violation count, and audit completeness.

Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.

- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.
- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.
- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

Acceptance Criteria

- Provider Ecosystem has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

Traceability

- MAL-V00-B009
- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B011
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

Version History

Version	Date	Status	Notes
v13 draft	2026-06-29	Draft	Initial Provider Ecosystem blueprint content for Mariam Architecture Library v13.

Book 03.11 - Provider Ecosystem

Version: v13 draft

Status: Draft

Document ID: MAL-V03-B03-11-INDEX

Purpose

Define the Provider Ecosystem blueprint package and its subsystem decomposition as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for the Provider Ecosystem blueprint package and its subsystem decomposition. It does not implement product code or bypass the documentation-first governance model.

Responsibilities

- Establish a canonical blueprint boundary for the Provider Ecosystem blueprint package and its subsystem decomposition.
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- Volume 02 Master Roadmap phase guidance corresponding to Provider Ecosystem.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

Outputs

- Blueprint contract and subsystem boundaries for the Provider Ecosystem blueprint package and its subsystem decomposition.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

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- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.
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- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

Acceptance Criteria

- Provider Ecosystem has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
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Traceability

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- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B011
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

Version History

Version	Date	Status	Notes
v13 draft	2026-06-29	Draft	Initial Provider Ecosystem blueprint content for Mariam Architecture Library v13.

book-03-11-01-provider-ecosystem-overview.md

Book 03.11.01 - Provider Ecosystem Overview

Version: v13 draft

Status: Draft

Document ID: MAL-V03-B03-11-001

Purpose

Define Provider Ecosystem Overview within the Provider Ecosystem blueprint as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for Provider Ecosystem Overview within the Provider Ecosystem blueprint. It does not implement product code or bypass the documentation-first governance model.

Responsibilities

- Establish a canonical blueprint boundary for Provider Ecosystem Overview within the Provider Ecosystem blueprint.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Provider Ecosystem.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

Outputs

- Blueprint contract and subsystem boundaries for Provider Ecosystem Overview within the Provider Ecosystem blueprint.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.
- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

Events

- ProviderEcosystemOverviewDefined
- ProviderEcosystemOverviewRegistered
- ProviderEcosystemOverviewStateChanged
- ProviderEcosystemOverviewReviewRequested
- ProviderEcosystemOverviewGovernanceBlocked

Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
- Retention metadata that separates draft, active, archived, deprecated, and superseded records.

Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
- Classify inputs, outputs, events, logs, metrics, memory, and exported artifacts before sharing or persistence.
- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

Metrics

- Adoption rate, successful completion rate, rejection rate, rework count, and user trust indicators.
- Latency, cost, throughput, availability, error rate, fallback rate, and recovery time.
- Quality score, benchmark pass rate, validation pass rate, review backlog, and stale-record count.
- Security denial count, governance exception count, policy violation count, and audit completeness.

Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.
- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.

- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.
- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

Acceptance Criteria

- Provider Ecosystem Overview has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

Traceability

- MAL-V00-B009
- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B011
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

Version History

Version	Date	Status	Notes
v13 draft	2026-06-29	Draft	Initial Provider Ecosystem blueprint content for Mariam Architecture Library v13.

Book 03.11.02 - Provider Registry

Version: v13 draft

Status: Draft

Document ID: MAL-V03-B03-11-002

Purpose

Define Provider Registry within the Provider Ecosystem blueprint as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for Provider Registry within the Provider Ecosystem blueprint. It does not implement product code or bypass the documentation-first governance model.

Responsibilities

- Establish a canonical blueprint boundary for Provider Registry within the Provider Ecosystem blueprint.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Provider Ecosystem.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

Outputs

- Blueprint contract and subsystem boundaries for Provider Registry within the Provider Ecosystem blueprint.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.

- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

Events

- ProviderRegistryDefined
- ProviderRegistryRegistered
- ProviderRegistryStateChanged
- ProviderRegistryReviewRequested
- ProviderRegistryGovernanceBlocked

Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
- Retention metadata that separates draft, active, archived, deprecated, and superseded records.

Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
- Classify inputs, outputs, events, logs, metrics, memory, and exported artifacts before sharing or persistence.
- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

Metrics

- Adoption rate, successful completion rate, rejection rate, rework count, and user trust indicators.
- Latency, cost, throughput, availability, error rate, fallback rate, and recovery time.
- Quality score, benchmark pass rate, validation pass rate, review backlog, and stale-record count.
- Security denial count, governance exception count, policy violation count, and audit completeness.

Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.
- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.
- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.

- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

Acceptance Criteria

- Provider Registry has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

Traceability

- MAL-V00-B009
- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B011
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

Version History

Version	Date	Status	Notes
v13 draft	2026-06-29	Draft	Initial Provider Ecosystem blueprint content for Mariam Architecture Library v13.

Book 03.11.03 - Provider Contracts

Version: v13 draft

Status: Draft

Document ID: MAL-V03-B03-11-003

Purpose

Define Provider Contracts within the Provider Ecosystem blueprint as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for Provider Contracts within the Provider Ecosystem blueprint. It does not implement product code or bypass the documentation-first governance model.

Responsibilities

- Establish a canonical blueprint boundary for Provider Contracts within the Provider Ecosystem blueprint.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Provider Ecosystem.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

Outputs

- Blueprint contract and subsystem boundaries for Provider Contracts within the Provider Ecosystem blueprint.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.

- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

Events

- ProviderContractsDefined
- ProviderContractsRegistered
- ProviderContractsStateChanged
- ProviderContractsReviewRequested
- ProviderContractsGovernanceBlocked

Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
- Retention metadata that separates draft, active, archived, deprecated, and superseded records.

Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
- Classify inputs, outputs, events, logs, metrics, memory, and exported artifacts before sharing or persistence.
- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

Metrics

- Adoption rate, successful completion rate, rejection rate, rework count, and user trust indicators.
- Latency, cost, throughput, availability, error rate, fallback rate, and recovery time.
- Quality score, benchmark pass rate, validation pass rate, review backlog, and stale-record count.
- Security denial count, governance exception count, policy violation count, and audit completeness.

Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.
- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.
- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.

- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

Acceptance Criteria

- Provider Contracts has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

Traceability

- MAL-V00-B009
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- MAL-V02-B011
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

Version History

Version	Date	Status	Notes
v13 draft	2026-06-29	Draft	Initial Provider Ecosystem blueprint content for Mariam Architecture Library v13.

Book 03.11.04 - Provider Routing

Version: v13 draft

Status: Draft

Document ID: MAL-V03-B03-11-004

Purpose

Define Provider Routing within the Provider Ecosystem blueprint as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for Provider Routing within the Provider Ecosystem blueprint. It does not implement product code or bypass the documentation-first governance model.

Responsibilities

- Establish a canonical blueprint boundary for Provider Routing within the Provider Ecosystem blueprint.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Provider Ecosystem.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

Outputs

- Blueprint contract and subsystem boundaries for Provider Routing within the Provider Ecosystem blueprint.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.

- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

Events

- ProviderRoutingDefined
- ProviderRoutingRegistered
- ProviderRoutingStateChanged
- ProviderRoutingReviewRequested
- ProviderRoutingGovernanceBlocked

Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
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- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

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- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
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- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.
- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.

- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

Acceptance Criteria

- Provider Routing has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

Traceability

- MAL-V00-B009
- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B011
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

Version History

Version	Date	Status	Notes
v13 draft	2026-06-29	Draft	Initial Provider Ecosystem blueprint content for Mariam Architecture Library v13.

Book 03.11.05 - Provider Credentials

Version: v13 draft

Status: Draft

Document ID: MAL-V03-B03-11-005

Purpose

Define Provider Credentials within the Provider Ecosystem blueprint as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for Provider Credentials within the Provider Ecosystem blueprint. It does not implement product code or bypass the documentation-first governance model.

Responsibilities

- Establish a canonical blueprint boundary for Provider Credentials within the Provider Ecosystem blueprint.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Provider Ecosystem.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

Outputs

- Blueprint contract and subsystem boundaries for Provider Credentials within the Provider Ecosystem blueprint.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.

- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

Events

- ProviderCredentialsDefined
- ProviderCredentialsRegistered
- ProviderCredentialsStateChanged
- ProviderCredentialsReviewRequested
- ProviderCredentialsGovernanceBlocked

Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
- Retention metadata that separates draft, active, archived, deprecated, and superseded records.

Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
- Classify inputs, outputs, events, logs, metrics, memory, and exported artifacts before sharing or persistence.
- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

Metrics

- Adoption rate, successful completion rate, rejection rate, rework count, and user trust indicators.
- Latency, cost, throughput, availability, error rate, fallback rate, and recovery time.
- Quality score, benchmark pass rate, validation pass rate, review backlog, and stale-record count.
- Security denial count, governance exception count, policy violation count, and audit completeness.

Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.
- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.
- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.

- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

Acceptance Criteria

- Provider Credentials has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

Traceability

- MAL-V00-B009
- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B011
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

Version History

Version	Date	Status	Notes
v13 draft	2026-06-29	Draft	Initial Provider Ecosystem blueprint content for Mariam Architecture Library v13.

book-03-11-06-provider-quotas.md

Book 03.11.06 - Provider Quotas

Version: v13 draft

Status: Draft

Document ID: MAL-V03-B03-11-006

Purpose

Define Provider Quotas within the Provider Ecosystem blueprint as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for Provider Quotas within the Provider Ecosystem blueprint. It does not implement product code or bypass the documentation-first governance model.

Responsibilities

- Establish a canonical blueprint boundary for Provider Quotas within the Provider Ecosystem blueprint.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Provider Ecosystem.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

Outputs

- Blueprint contract and subsystem boundaries for Provider Quotas within the Provider Ecosystem blueprint.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.

- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

Events

- ProviderQuotasDefined
- ProviderQuotasRegistered
- ProviderQuotasStateChanged
- ProviderQuotasReviewRequested
- ProviderQuotasGovernanceBlocked

Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
- Retention metadata that separates draft, active, archived, deprecated, and superseded records.

Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
- Classify inputs, outputs, events, logs, metrics, memory, and exported artifacts before sharing or persistence.
- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

Metrics

- Adoption rate, successful completion rate, rejection rate, rework count, and user trust indicators.
- Latency, cost, throughput, availability, error rate, fallback rate, and recovery time.
- Quality score, benchmark pass rate, validation pass rate, review backlog, and stale-record count.
- Security denial count, governance exception count, policy violation count, and audit completeness.

Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.
- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.
- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.

- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

Acceptance Criteria

- Provider Quotas has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

Traceability

- MAL-V00-B009
- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B011
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

Version History

Version	Date	Status	Notes
v13 draft	2026-06-29	Draft	Initial Provider Ecosystem blueprint content for Mariam Architecture Library v13.

Book 03.11.07 - Provider Health

Version: v13 draft

Status: Draft

Document ID: MAL-V03-B03-11-007

Purpose

Define Provider Health within the Provider Ecosystem blueprint as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for Provider Health within the Provider Ecosystem blueprint. It does not implement product code or bypass the documentation-first governance model.

Responsibilities

- Establish a canonical blueprint boundary for Provider Health within the Provider Ecosystem blueprint.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Provider Ecosystem.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

Outputs

- Blueprint contract and subsystem boundaries for Provider Health within the Provider Ecosystem blueprint.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.

- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

Events

- ProviderHealthDefined
- ProviderHealthRegistered
- ProviderHealthStateChanged
- ProviderHealthReviewRequested
- ProviderHealthGovernanceBlocked

Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
- Retention metadata that separates draft, active, archived, deprecated, and superseded records.

Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
- Classify inputs, outputs, events, logs, metrics, memory, and exported artifacts before sharing or persistence.
- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

Metrics

- Adoption rate, successful completion rate, rejection rate, rework count, and user trust indicators.
- Latency, cost, throughput, availability, error rate, fallback rate, and recovery time.
- Quality score, benchmark pass rate, validation pass rate, review backlog, and stale-record count.
- Security denial count, governance exception count, policy violation count, and audit completeness.

Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.
- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.
- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.

- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

Acceptance Criteria

- Provider Health has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

Traceability

- MAL-V00-B009
- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B011
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

Version History

Version	Date	Status	Notes
v13 draft	2026-06-29	Draft	Initial Provider Ecosystem blueprint content for Mariam Architecture Library v13.

Book 03.11.08 - Provider Fallback

Version: v13 draft

Status: Draft

Document ID: MAL-V03-B03-11-008

Purpose

Define Provider Fallback within the Provider Ecosystem blueprint as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for Provider Fallback within the Provider Ecosystem blueprint. It does not implement product code or bypass the documentation-first governance model.

Responsibilities

- Establish a canonical blueprint boundary for Provider Fallback within the Provider Ecosystem blueprint.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Provider Ecosystem.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

Outputs

- Blueprint contract and subsystem boundaries for Provider Fallback within the Provider Ecosystem blueprint.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.

- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

Events

- ProviderFallbackDefined
- ProviderFallbackRegistered
- ProviderFallbackStateChanged
- ProviderFallbackReviewRequested
- ProviderFallbackGovernanceBlocked

Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
- Retention metadata that separates draft, active, archived, deprecated, and superseded records.

Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
- Classify inputs, outputs, events, logs, metrics, memory, and exported artifacts before sharing or persistence.
- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

Metrics

- Adoption rate, successful completion rate, rejection rate, rework count, and user trust indicators.
- Latency, cost, throughput, availability, error rate, fallback rate, and recovery time.
- Quality score, benchmark pass rate, validation pass rate, review backlog, and stale-record count.
- Security denial count, governance exception count, policy violation count, and audit completeness.

Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.
- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.
- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.

- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

Acceptance Criteria

- Provider Fallback has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

Traceability

- MAL-V00-B009
- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B011
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

Version History

Version	Date	Status	Notes
v13 draft	2026-06-29	Draft	Initial Provider Ecosystem blueprint content for Mariam Architecture Library v13.

Book 03.11.09 - Provider Adapters

Version: v13 draft

Status: Draft

Document ID: MAL-V03-B03-11-009

Purpose

Define Provider Adapters within the Provider Ecosystem blueprint as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for Provider Adapters within the Provider Ecosystem blueprint. It does not implement product code or bypass the documentation-first governance model.

Responsibilities

- Establish a canonical blueprint boundary for Provider Adapters within the Provider Ecosystem blueprint.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Provider Ecosystem.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

Outputs

- Blueprint contract and subsystem boundaries for Provider Adapters within the Provider Ecosystem blueprint.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.

- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

Events

- ProviderAdaptersDefined
- ProviderAdaptersRegistered
- ProviderAdaptersStateChanged
- ProviderAdaptersReviewRequested
- ProviderAdaptersGovernanceBlocked

Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
- Retention metadata that separates draft, active, archived, deprecated, and superseded records.

Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
- Classify inputs, outputs, events, logs, metrics, memory, and exported artifacts before sharing or persistence.
- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

Metrics

- Adoption rate, successful completion rate, rejection rate, rework count, and user trust indicators.
- Latency, cost, throughput, availability, error rate, fallback rate, and recovery time.
- Quality score, benchmark pass rate, validation pass rate, review backlog, and stale-record count.
- Security denial count, governance exception count, policy violation count, and audit completeness.

Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.
- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.
- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.

- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

Acceptance Criteria

- Provider Adapters has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

Traceability

- MAL-V00-B009
- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B011
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

Version History

Version	Date	Status	Notes
v13 draft	2026-06-29	Draft	Initial Provider Ecosystem blueprint content for Mariam Architecture Library v13.

Book 03.11.10 - Provider Billing

Version: v13 draft

Status: Draft

Document ID: MAL-V03-B03-11-010

Purpose

Define Provider Billing within the Provider Ecosystem blueprint as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for Provider Billing within the Provider Ecosystem blueprint. It does not implement product code or bypass the documentation-first governance model.

Responsibilities

- Establish a canonical blueprint boundary for Provider Billing within the Provider Ecosystem blueprint.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Provider Ecosystem.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

Outputs

- Blueprint contract and subsystem boundaries for Provider Billing within the Provider Ecosystem blueprint.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.

- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

Events

- ProviderBillingDefined
- ProviderBillingRegistered
- ProviderBillingStateChanged
- ProviderBillingReviewRequested
- ProviderBillingGovernanceBlocked

Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
- Retention metadata that separates draft, active, archived, deprecated, and superseded records.

Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
- Classify inputs, outputs, events, logs, metrics, memory, and exported artifacts before sharing or persistence.
- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

Metrics

- Adoption rate, successful completion rate, rejection rate, rework count, and user trust indicators.
- Latency, cost, throughput, availability, error rate, fallback rate, and recovery time.
- Quality score, benchmark pass rate, validation pass rate, review backlog, and stale-record count.
- Security denial count, governance exception count, policy violation count, and audit completeness.

Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.
- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.
- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.

- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

Acceptance Criteria

- Provider Billing has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

Traceability

- MAL-V00-B009
- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B011
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

Version History

Version	Date	Status	Notes
v13 draft	2026-06-29	Draft	Initial Provider Ecosystem blueprint content for Mariam Architecture Library v13.

Book 03.11.11 - Provider Compliance

Version: v13 draft

Status: Draft

Document ID: MAL-V03-B03-11-011

Purpose

Define Provider Compliance within the Provider Ecosystem blueprint as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for Provider Compliance within the Provider Ecosystem blueprint. It does not implement product code or bypass the documentation-first governance model.

Responsibilities

- Establish a canonical blueprint boundary for Provider Compliance within the Provider Ecosystem blueprint.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Provider Ecosystem.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

Outputs

- Blueprint contract and subsystem boundaries for Provider Compliance within the Provider Ecosystem blueprint.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.

- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

Events

- ProviderComplianceDefined
- ProviderComplianceRegistered
- ProviderComplianceStateChanged
- ProviderComplianceReviewRequested
- ProviderComplianceGovernanceBlocked

Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
- Retention metadata that separates draft, active, archived, deprecated, and superseded records.

Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
- Classify inputs, outputs, events, logs, metrics, memory, and exported artifacts before sharing or persistence.
- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

Metrics

- Adoption rate, successful completion rate, rejection rate, rework count, and user trust indicators.
- Latency, cost, throughput, availability, error rate, fallback rate, and recovery time.
- Quality score, benchmark pass rate, validation pass rate, review backlog, and stale-record count.
- Security denial count, governance exception count, policy violation count, and audit completeness.

Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.
- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.
- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.

- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

Acceptance Criteria

- Provider Compliance has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

Traceability

- MAL-V00-B009
- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B011
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

Version History

Version	Date	Status	Notes
v13 draft	2026-06-29	Draft	Initial Provider Ecosystem blueprint content for Mariam Architecture Library v13.

Book 03.11.12 - Provider Onboarding

Version: v13 draft

Status: Draft

Document ID: MAL-V03-B03-11-012

Purpose

Define Provider Onboarding within the Provider Ecosystem blueprint as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for Provider Onboarding within the Provider Ecosystem blueprint. It does not implement product code or bypass the documentation-first governance model.

Responsibilities

- Establish a canonical blueprint boundary for Provider Onboarding within the Provider Ecosystem blueprint.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Provider Ecosystem.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

Outputs

- Blueprint contract and subsystem boundaries for Provider Onboarding within the Provider Ecosystem blueprint.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.

- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

Events

- ProviderOnboardingDefined
- ProviderOnboardingRegistered
- ProviderOnboardingStateChanged
- ProviderOnboardingReviewRequested
- ProviderOnboardingGovernanceBlocked

Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
- Retention metadata that separates draft, active, archived, deprecated, and superseded records.

Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
- Classify inputs, outputs, events, logs, metrics, memory, and exported artifacts before sharing or persistence.
- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

Metrics

- Adoption rate, successful completion rate, rejection rate, rework count, and user trust indicators.
- Latency, cost, throughput, availability, error rate, fallback rate, and recovery time.
- Quality score, benchmark pass rate, validation pass rate, review backlog, and stale-record count.
- Security denial count, governance exception count, policy violation count, and audit completeness.

Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.
- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.
- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.

- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

Acceptance Criteria

- Provider Onboarding has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

Traceability

- MAL-V00-B009
- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B011
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

Version History

Version	Date	Status	Notes
v13 draft	2026-06-29	Draft	Initial Provider Ecosystem blueprint content for Mariam Architecture Library v13.

Book 03.11.13 - Provider Deprecation

Version: v13 draft

Status: Draft

Document ID: MAL-V03-B03-11-013

Purpose

Define Provider Deprecation within the Provider Ecosystem blueprint as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for Provider Deprecation within the Provider Ecosystem blueprint. It does not implement product code or bypass the documentation-first governance model.

Responsibilities

- Establish a canonical blueprint boundary for Provider Deprecation within the Provider Ecosystem blueprint.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Provider Ecosystem.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

Outputs

- Blueprint contract and subsystem boundaries for Provider Deprecation within the Provider Ecosystem blueprint.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.

- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

Events

- ProviderDeprecationDefined
- ProviderDeprecationRegistered
- ProviderDeprecationStateChanged
- ProviderDeprecationReviewRequested
- ProviderDeprecationGovernanceBlocked

Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
- Retention metadata that separates draft, active, archived, deprecated, and superseded records.

Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
- Classify inputs, outputs, events, logs, metrics, memory, and exported artifacts before sharing or persistence.
- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

Metrics

- Adoption rate, successful completion rate, rejection rate, rework count, and user trust indicators.
- Latency, cost, throughput, availability, error rate, fallback rate, and recovery time.
- Quality score, benchmark pass rate, validation pass rate, review backlog, and stale-record count.
- Security denial count, governance exception count, policy violation count, and audit completeness.

Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.
- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.
- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.

- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

Acceptance Criteria

- Provider Deprecation has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

Traceability

- MAL-V00-B009
- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B011
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

Version History

Version	Date	Status	Notes
v13 draft	2026-06-29	Draft	Initial Provider Ecosystem blueprint content for Mariam Architecture Library v13.

Book 03.11.14 - Provider Observability

Version: v13 draft

Status: Draft

Document ID: MAL-V03-B03-11-014

Purpose

Define Provider Observability within the Provider Ecosystem blueprint as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for Provider Observability within the Provider Ecosystem blueprint. It does not implement product code or bypass the documentation-first governance model.

Responsibilities

- Establish a canonical blueprint boundary for Provider Observability within the Provider Ecosystem blueprint.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Provider Ecosystem.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

Outputs

- Blueprint contract and subsystem boundaries for Provider Observability within the Provider Ecosystem blueprint.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.

- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

Events

- ProviderObservabilityDefined
- ProviderObservabilityRegistered
- ProviderObservabilityStateChanged
- ProviderObservabilityReviewRequested
- ProviderObservabilityGovernanceBlocked

Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
- Retention metadata that separates draft, active, archived, deprecated, and superseded records.

Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
- Classify inputs, outputs, events, logs, metrics, memory, and exported artifacts before sharing or persistence.
- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

Metrics

- Adoption rate, successful completion rate, rejection rate, rework count, and user trust indicators.
- Latency, cost, throughput, availability, error rate, fallback rate, and recovery time.
- Quality score, benchmark pass rate, validation pass rate, review backlog, and stale-record count.
- Security denial count, governance exception count, policy violation count, and audit completeness.

Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.
- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.
- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.

- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

Acceptance Criteria

- Provider Observability has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

Traceability

- MAL-V00-B009
- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B011
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

Version History

Version	Date	Status	Notes
v13 draft	2026-06-29	Draft	Initial Provider Ecosystem blueprint content for Mariam Architecture Library v13.

Book 03.11.15 - Provider Governance

Version: v13 draft

Status: Draft

Document ID: MAL-V03-B03-11-015

Purpose

Define Provider Governance within the Provider Ecosystem blueprint as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for Provider Governance within the Provider Ecosystem blueprint. It does not implement product code or bypass the documentation-first governance model.

Responsibilities

- Establish a canonical blueprint boundary for Provider Governance within the Provider Ecosystem blueprint.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Provider Ecosystem.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

Outputs

- Blueprint contract and subsystem boundaries for Provider Governance within the Provider Ecosystem blueprint.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.

- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

Events

- ProviderGovernanceDefined
- ProviderGovernanceRegistered
- ProviderGovernanceStateChanged
- ProviderGovernanceReviewRequested
- ProviderGovernanceGovernanceBlocked

Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
- Retention metadata that separates draft, active, archived, deprecated, and superseded records.

Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
- Classify inputs, outputs, events, logs, metrics, memory, and exported artifacts before sharing or persistence.
- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

Metrics

- Adoption rate, successful completion rate, rejection rate, rework count, and user trust indicators.
- Latency, cost, throughput, availability, error rate, fallback rate, and recovery time.
- Quality score, benchmark pass rate, validation pass rate, review backlog, and stale-record count.
- Security denial count, governance exception count, policy violation count, and audit completeness.

Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.
- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.
- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.

- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

Acceptance Criteria

- Provider Governance has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

Traceability

- MAL-V00-B009
- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B011
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

Version History

Version	Date	Status	Notes
v13 draft	2026-06-29	Draft	Initial Provider Ecosystem blueprint content for Mariam Architecture Library v13.

Book 03.11.16 - Provider Events

Version: v13 draft

Status: Draft

Document ID: MAL-V03-B03-11-016

Purpose

Define Provider Events within the Provider Ecosystem blueprint as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for Provider Events within the Provider Ecosystem blueprint. It does not implement product code or bypass the documentation-first governance model.

Responsibilities

- Establish a canonical blueprint boundary for Provider Events within the Provider Ecosystem blueprint.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Provider Ecosystem.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

Outputs

- Blueprint contract and subsystem boundaries for Provider Events within the Provider Ecosystem blueprint.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.

- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

Events

- ProviderEventsDefined
- ProviderEventsRegistered
- ProviderEventsStateChanged
- ProviderEventsReviewRequested
- ProviderEventsGovernanceBlocked

Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
- Retention metadata that separates draft, active, archived, deprecated, and superseded records.

Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
- Classify inputs, outputs, events, logs, metrics, memory, and exported artifacts before sharing or persistence.
- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

Metrics

- Adoption rate, successful completion rate, rejection rate, rework count, and user trust indicators.
- Latency, cost, throughput, availability, error rate, fallback rate, and recovery time.
- Quality score, benchmark pass rate, validation pass rate, review backlog, and stale-record count.
- Security denial count, governance exception count, policy violation count, and audit completeness.

Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.
- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.
- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.

- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

Acceptance Criteria

- Provider Events has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

Traceability

- MAL-V00-B009
- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B011
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

Version History

Version	Date	Status	Notes
v13 draft	2026-06-29	Draft	Initial Provider Ecosystem blueprint content for Mariam Architecture Library v13.

book-03-11-17-provider-storage.md

Book 03.11.17 - Provider Storage

Version: v13 draft

Status: Draft

Document ID: MAL-V03-B03-11-017

Purpose

Define Provider Storage within the Provider Ecosystem blueprint as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for Provider Storage within the Provider Ecosystem blueprint. It does not implement product code or bypass the documentation-first governance model.

Responsibilities

- Establish a canonical blueprint boundary for Provider Storage within the Provider Ecosystem blueprint.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Provider Ecosystem.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

Outputs

- Blueprint contract and subsystem boundaries for Provider Storage within the Provider Ecosystem blueprint.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.

- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

Events

- ProviderStorageDefined
- ProviderStorageRegistered
- ProviderStorageStateChanged
- ProviderStorageReviewRequested
- ProviderStorageGovernanceBlocked

Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
- Retention metadata that separates draft, active, archived, deprecated, and superseded records.

Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
- Classify inputs, outputs, events, logs, metrics, memory, and exported artifacts before sharing or persistence.
- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

Metrics

- Adoption rate, successful completion rate, rejection rate, rework count, and user trust indicators.
- Latency, cost, throughput, availability, error rate, fallback rate, and recovery time.
- Quality score, benchmark pass rate, validation pass rate, review backlog, and stale-record count.
- Security denial count, governance exception count, policy violation count, and audit completeness.

Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.
- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.
- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.

- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

Acceptance Criteria

- Provider Storage has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

Traceability

- MAL-V00-B009
- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B011
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

Version History

Version	Date	Status	Notes
v13 draft	2026-06-29	Draft	Initial Provider Ecosystem blueprint content for Mariam Architecture Library v13.

Book 03.11.18 - Provider Security

Version: v13 draft

Status: Draft

Document ID: MAL-V03-B03-11-018

Purpose

Define Provider Security within the Provider Ecosystem blueprint as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for Provider Security within the Provider Ecosystem blueprint. It does not implement product code or bypass the documentation-first governance model.

Responsibilities

- Establish a canonical blueprint boundary for Provider Security within the Provider Ecosystem blueprint.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Provider Ecosystem.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

Outputs

- Blueprint contract and subsystem boundaries for Provider Security within the Provider Ecosystem blueprint.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.

- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

Events

- ProviderSecurityDefined
- ProviderSecurityRegistered
- ProviderSecurityStateChanged
- ProviderSecurityReviewRequested
- ProviderSecurityGovernanceBlocked

Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
- Retention metadata that separates draft, active, archived, deprecated, and superseded records.

Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
- Classify inputs, outputs, events, logs, metrics, memory, and exported artifacts before sharing or persistence.
- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

Metrics

- Adoption rate, successful completion rate, rejection rate, rework count, and user trust indicators.
- Latency, cost, throughput, availability, error rate, fallback rate, and recovery time.
- Quality score, benchmark pass rate, validation pass rate, review backlog, and stale-record count.
- Security denial count, governance exception count, policy violation count, and audit completeness.

Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.
- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.
- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.

- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

Acceptance Criteria

- Provider Security has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

Traceability

- MAL-V00-B009
- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B011
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

Version History

Version	Date	Status	Notes
v13 draft	2026-06-29	Draft	Initial Provider Ecosystem blueprint content for Mariam Architecture Library v13.

book-03-11-19-provider-testing.md

Book 03.11.19 - Provider Testing

Version: v13 draft

Status: Draft

Document ID: MAL-V03-B03-11-019

Purpose

Define Provider Testing within the Provider Ecosystem blueprint as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for Provider Testing within the Provider Ecosystem blueprint. It does not implement product code or bypass the documentation-first governance model.

Responsibilities

- Establish a canonical blueprint boundary for Provider Testing within the Provider Ecosystem blueprint.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Provider Ecosystem.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

Outputs

- Blueprint contract and subsystem boundaries for Provider Testing within the Provider Ecosystem blueprint.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.

- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

Events

- ProviderTestingDefined
- ProviderTestingRegistered
- ProviderTestingStateChanged
- ProviderTestingReviewRequested
- ProviderTestingGovernanceBlocked

Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
- Retention metadata that separates draft, active, archived, deprecated, and superseded records.

Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
- Classify inputs, outputs, events, logs, metrics, memory, and exported artifacts before sharing or persistence.
- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

Metrics

- Adoption rate, successful completion rate, rejection rate, rework count, and user trust indicators.
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Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.
- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.
- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.

- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

Acceptance Criteria

- Provider Testing has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

Traceability

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- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B011
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

Version History

Version	Date	Status	Notes
v13 draft	2026-06-29	Draft	Initial Provider Ecosystem blueprint content for Mariam Architecture Library v13.

book-03-11-20-provider-roadmap.md

Book 03.11.20 - Provider Roadmap

Version: v13 draft

Status: Draft

Document ID: MAL-V03-B03-11-020

Purpose

Define Provider Roadmap within the Provider Ecosystem blueprint as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for Provider Roadmap within the Provider Ecosystem blueprint. It does not implement product code or bypass the documentation-first governance model.

Responsibilities

- Establish a canonical blueprint boundary for Provider Roadmap within the Provider Ecosystem blueprint.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Provider Ecosystem.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

Outputs

- Blueprint contract and subsystem boundaries for Provider Roadmap within the Provider Ecosystem blueprint.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.

- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
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- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

Events

- ProviderRoadmapDefined
- ProviderRoadmapRegistered
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- ProviderRoadmapReviewRequested
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Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
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Acceptance Criteria

- Provider Roadmap has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
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Version History

Version	Date	Status	Notes
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